Pavel Golikov

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PROFESSIONAL PROFILE

AI Researcher and agent engineer specializing in robust machine reasoning, retrieval-augmented generation (RAG), LLM evaluation, and alignment. Builds grounded agentic workflows, vector-retrieval systems, and dataset generators for context-management evaluation in LLM agents. Former Military Intelligence Operator who held a Top Secret clearance, bringing a threat-modeling mindset to AI security and model capability evaluation.

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, LaTeX
AI & Agents: PyTorch, Reinforcement Learning (RL), RLVR, LangGraph, LangChain, RAG
Hugging Face, vLLM, Transformers, Gemini/OpenAI/Anthropic APIs
RAG & Evals: Vector storage, embeddings, BM25, hybrid retrieval, cross-encoder reranking, MRR/nDCG/recall, groundedness, validation
Systems: Linux, SQLite, GitHub Actions CI, AWS, Apache Flink, Distributed GPU Clusters

AI RESEARCH & AGENT ENGINEERING

- **ArbiGraph** | *arXiv* | *GitHub* | *Hugging Face* — 2026 – Present
  - Built an open-source benchmark and dataset generator for evaluating context management in tool-assisted LLM agents across arbitrarily scalable, long-horizon task graphs.
  - Generated math, word-problem, and Python-tracing workflows with executable ground truth, enabling exact evaluation of state retention, updates, propagation, and stale-state reuse without manual labels.
  - Built reinforcement learning with verifiable rewards (RLVR) environments supporting dataset-backed and on-demand episodes, hidden verifier state, exact binary rewards, and per-node diagnostics.
- **arXiv Research Agent** | *GitHub* | *Evals* — 2026
  - Built an evaluation-first **RAG** literature-review agent with **LangGraph, LangChain, Gemini, and Chroma vector storage**; orchestrates arXiv search, PDF parsing, hybrid retrieval, reranking, and citation-grounded synthesis.
  - Benchmarked dense, BM25, hybrid, and cross-encoder retrieval on 50 hand-labeled questions over 570 chunks using MRR, nDCG, recall, and paired-bootstrap confidence intervals; evaluated groundedness and claim support.
  - Engineered deterministic map-reduce fan-out, retry-aware partial results, and resumable **SQLite checkpointing**; shipped 163 offline tests in CI and measured a median end-to-end cost of $0.054 per run.
- **Robust Reasoning Benchmark (RRB)** | *arXiv* | *DOI* | *GitHub* — 2026
  - Designed an adversarial evaluation framework leveraging 13 deterministic textual perturbations to decouple an LLM's mechanical deciphering from its underlying mathematical logic.
  - Demonstrated that the attention drift occurs *within a single query's Chain-of-Thought*, empirically showing that intermediate reasoning steps pollute the dense attention mechanism.
  - Engineered custom mechanistic interpretability pipelines in **PyTorch** for layerwise attention-allocation analysis across token index boundaries, testing models ranging from 7B to 30B parameters.
- **Fusing Adds and Shifts for Efficient Dot Products** | *IEEE CAL* | GitHub — 2026
  Hardware ML Research — *Toronto, ON*
  - Proposed and validated a novel algorithmic optimization for dot-product computations, demonstrating a strong foundational understanding of hardware-level ML primitives and efficiency.

ENGINEERING & PROFESSIONAL EXPERIENCE

- **Graduate Researcher (PhD, on leave) — ML Systems & Agent Evaluation** — 2022 – Present
  University of Toronto, EcoSystem Research Group — *Toronto, ON*
  - Conducted research spanning ML systems and efficient computation, with later work focused on robust machine reasoning, agent evaluation, context management, and AI alignment, in the EcoSystem Research Group and as a member of the Vector Institute.
  - Developed open-source evaluation frameworks, agentic RAG systems, and mechanistic-interpretability analyses, resulting in first-author research on context management and reasoning robustness.
- **Graduate Researcher (MSc) — Distributed Systems** — 2020 – 2022
  University of Toronto, EcoSystem Research Group — *Toronto, ON*
  - Engineered a flexible IoT distributed data-streaming framework from scratch, designed to automatically partition computational streaming queries between edge devices and cloud instances.
  - Built the full software stack: programmed Arduino/C++ sensors for real-time biological data collection (EMG/ECG), developed custom socket networking protocols, and deployed cloud infrastructure using **AWS** and **Apache Flink**.
- **Intelligence Operator** — 2013 – 2018
  Canadian Armed Forces — *Canada*
  - Formerly held a **Top Secret** security clearance while conducting rigorous analysis of classified information streams to produce actionable intelligence reports for command elements.
  - Developed a strong adversarial threat-modeling mindset, emphasizing operational security, rigorous data validation, and the identification of logical vulnerabilities in complex, multi-agent scenarios.
- **Mathematics Teacher** — 2012 – 2013 & 2018 – 2019
  Blyth Academy — *Toronto*
  - Taught foundational mathematics to students in Grades 10, 11, and 12, developing the ability to distill and communicate complex quantitative concepts.

EDUCATION

- **University of Toronto** — Toronto, ON
  PhD in Computer Science (on leave as of September 2026) — *2022 – Present*
- **University of Toronto** — Toronto, ON
  Master of Science (MSc) in Computer Science — *2020 – 2022*
- **University of Toronto** — Toronto, ON
  Bachelor of Science (BSc) in Mathematics and Philosophy (Formal Logic) — *Graduated 2011*